

Static Analysis and Verification

Temporal Logics

Tomáš Vojnar, Ondřej Lengál

SAV 2026, FIT BUT

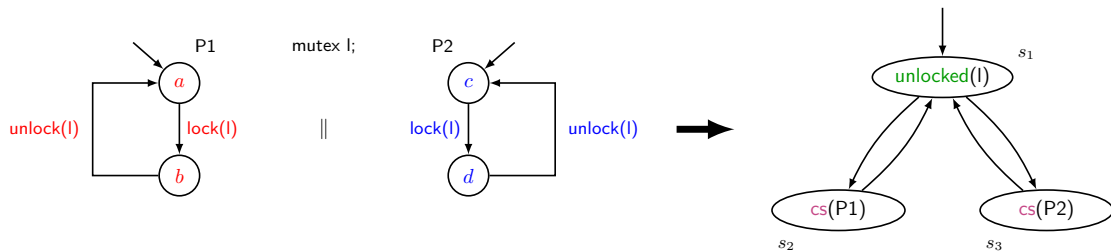
Temporal Logics

CTL*, CTL, LTL

Model of Computation

Kripke Structures

- Informally, **Kripke structures** are directed graphs whose
 - vertices** correspond to configurations of the examined system,
 - the vertices are **labelled** by atomic propositions that are true in the appropriate configurations, and
 - edges** encode possible transitions between the configurations.



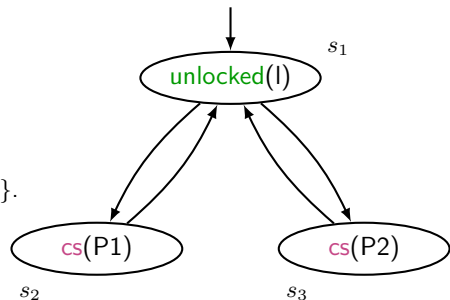
- Can be **generated** from the source description of examined systems (or used implicitly as an underlying semantic model of the formulae as well as examined systems).
- The generation involves the **state explosion problem**, or the Kripke structure may be **infinite**—in the following, we, however, concentrate on finite Kripke structures.

Kripke Structures

- Let AP be a set of **atomic propositions** about the configurations of the examined system.
- Formally, a (finite) **Kripke structure** M over AP is a tuple $M = (S, S_0, R, L)$ where
 - ▶ S is a finite set of **states**,
 - ▶ $S_0 \subseteq S$ is a set of **initial states**,
 - ▶ $R \subseteq S \times S$ is a **transition relation**, for convenience supposed to be total (i.e., $\forall s \in S \exists s' \in S: R(s, s')$),
 - ▶ $L: S \rightarrow 2^{AP}$ is a **labelling function** that labels each state by the set of atomic propositions that are true in it.

- For the example from the previous slide, we have:

- ▶ $AP = \{unlocked(l), cs(P1), cs(P2)\}$,
- ▶ $S = \{s_1, s_2, s_3\}$,
- ▶ $S_0 = \{s_1\}$,
- ▶ $R = \{(s_1, s_2), (s_2, s_1), (s_1, s_3), (s_3, s_1)\}$,
- ▶ $L = \{s_1 \mapsto \{unlocked(l)\}, s_2 \mapsto \{cs(P1)\}, s_3 \mapsto \{cs(P2)\}\}$.



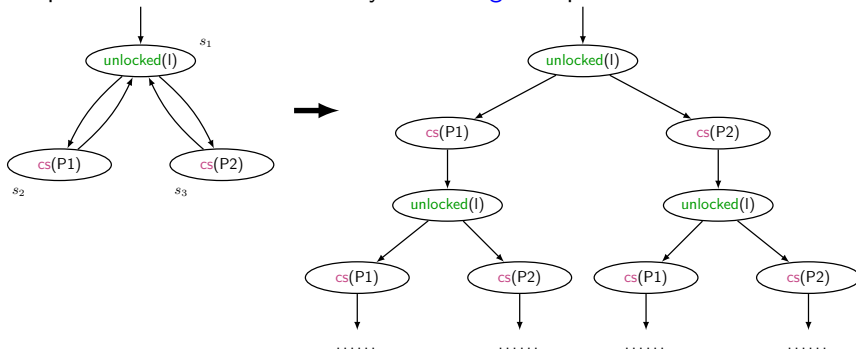
Kripke Structures

- A **path** π in a Kripke structure M is an infinite sequence of states $\pi = s_0 s_1 s_2 \dots$ such that $\forall i \in \mathbb{N}: R(s_i, s_{i+1})$.
- We denote $\Pi(M, s)$ the set of all paths in M that start at $s \in S$.
- The **suffix** π^i of a path $\pi = s_0 s_1 s_2 \dots s_i s_{i+1} s_{i+2} \dots$ is the path $\pi^i = s_i s_{i+1} s_{i+2} \dots$ starting at s_i .

The CTL* Logic

CTL* — Basic Idea

- CTL* formulae describe properties of **computation trees**.
- Infinite computation trees are obtained by **unwinding** a Kripke structure from its initial states.



- CTL* formulae **consist of**:
 - ▶ atomic propositions (= Boolean variables),
 - ▶ Boolean connectives,
 - ▶ **path quantifiers**,
 - ▶ **temporal operators**.

CTL* — Quantifiers and Operators

- **Path quantifiers** — describe the branching structure of a computation tree:
 - ▶ E : for some computation path leading from a state,
 - ▶ A : for all computation paths leading from a state.
- **Temporal operators** — properties of a path through a computation tree:
 - ▶ $X \varphi$ (“next time”, $\bigcirc$): the property φ holds (on the path starting) from the second state of the given path,
 - ▶ $F \varphi$ (“eventually” / “sometimes” (/ “finally”), $\Diamond$): the property φ holds (on the path starting) from some state of the given path,
 - ▶ $G \varphi$ (“always” / “globally”, $\Box$): the property φ holds from all states of the path,
 - ▶ $\varphi U \psi$ (“until”): the property ψ holds from some state of the path, and the property φ holds from all preceding states of the path,
 - ▶ $\varphi R \psi$ (“release”): the property ψ holds from all states of the path up to (and including) the first state from where the property φ holds (if such a state exists).

CTL* — Syntax

- Let AP be a non-empty set of atomic propositions.
- The syntax of **state formulae**, which are true in a specific state, is given by the following rules:
 - ▶ If $p \in AP$, then p is a state formula.
 - ▶ If φ and ψ are state formulae, then $\neg\varphi$, $\varphi \vee \psi$, $\varphi \wedge \psi$ are state formulae.
 - ▶ If φ is a path formula, then $E\varphi$ and $A\varphi$ are state formulae.
- The syntax of **path formulae**, which are true along a specific path, is given by the following rules:
 - ▶ If φ is a state formula, then φ is a path formula too.
 - ▶ If φ and ψ are path formulae, then $\neg\varphi$, $\varphi \vee \psi$, $\varphi \wedge \psi$, $X\varphi$, $F\varphi$, $G\varphi$, $\varphi U \psi$, and $\varphi R \psi$ are path formulae.
- CTL* is the set of **state formulae** generated by the above rules.

CTL* — Semantics

- Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .
- Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:
 - ▶ $M, s \models p$ iff $p \in L(s)$.
 - ▶ $M, s \models \neg\varphi_1$ iff $M, s \not\models \varphi_1$.
 - ▶ $M, s \models \varphi_1 \vee \varphi_2$ iff $M, s \models \varphi_1$ or $M, s \models \varphi_2$.
 - ▶ $M, s \models \varphi_1 \wedge \varphi_2$ iff $M, s \models \varphi_1$ and $M, s \models \varphi_2$.
 - ▶ $M, s \models E \psi_1$ iff $\exists \pi \in \Pi(M, s): M, \pi \models \psi_1$.
 - ▶ $M, s \models A \psi_1$ iff $\forall \pi \in \Pi(M, s): M, \pi \models \psi_1$.

Continued at the next slide...

■ Continued from the previous slide. . .

- ▶ $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- ▶ $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- ▶ $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.
- ▶ $M, \pi \models \psi_1 \wedge \psi_2$ iff $M, \pi \models \psi_1$ and $M, \pi \models \psi_2$.
- ▶ $M, \pi \models X \psi_1$ iff $M, \pi^1 \models \psi_1$.
- ▶ $M, \pi \models F \psi_1$ iff $\exists i \geq 0: M, \pi^i \models \psi_1$.
- ▶ $M, \pi \models G \psi_1$ iff $\forall i \geq 0: M, \pi^i \models \psi_1$.
- ▶ $M, \pi \models \psi_1 U \psi_2$ iff $\exists i \geq 0: M, \pi^i \models \psi_2$ and $\forall 0 \leq j < i: M, \pi^j \models \psi_1$.
- ▶ $M, \pi \models \psi_1 R \psi_2$ iff $\forall i \geq 0: (\forall 0 \leq j < i: M, \pi^j \not\models \psi_1) \Rightarrow M, \pi^i \models \psi_2$.
- ▶ For a (state) CTL* formula φ , we write $M \models \varphi$ iff $\forall s_0 \in S_0: M, s_0 \models \varphi$.

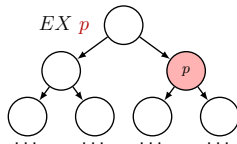
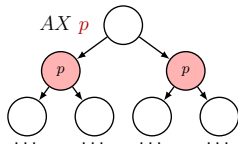
CTL* — Basic Operators

- Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :
 - ▶ let $p \in AP$, $true \equiv p \vee \neg p$ (and $false \equiv \neg true$),
 - ▶ $\varphi \wedge \psi \equiv \neg(\neg\varphi \vee \neg\psi)$,
 - ▶ $F \varphi \equiv true U \varphi$,
 - ▶ $G \varphi \equiv \neg F \neg\varphi$,
 - ▶ $\varphi R \psi \equiv \neg(\neg\varphi U \neg\psi)$,
 - ▶ $A \varphi \equiv \neg E \neg\varphi$.
- Some further connectives may be introduced too, e.g.:
 - ▶ $\varphi \rightarrow \psi \equiv \neg\varphi \vee \psi$,
 - ▶ $\varphi \leftrightarrow \psi \equiv (\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$,
 - ▶ ...

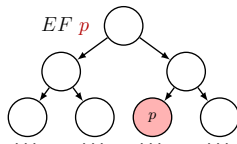
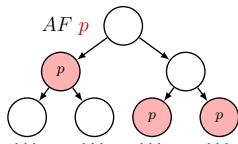
The CTL Logic

CTL — Syntax

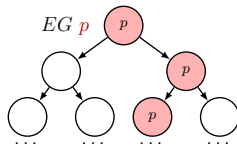
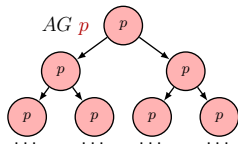
- CTL is a sublogic of CTL*—the path formulae are restricted to $X \varphi$, $F \varphi$, $G \varphi$, $\varphi U \psi$, and $\varphi R \psi$ for φ, ψ being state formulae.
- In effect, there are allowed these 10 modal CTL operators:
 - ▶ AX and EX ,



- ▶ AF and EF ,



- ▶ AG and EG ,

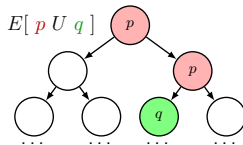
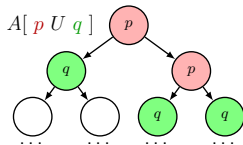


Continued at the next slide...

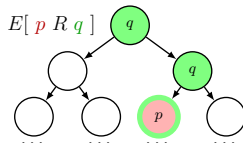
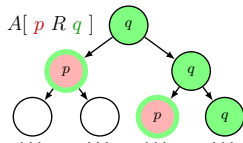
CTL — Syntax

- Continued from the previous slide...

- AU and EU ,



- AR and ER .



- There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \varphi \equiv \neg EX \neg \varphi$,

- $EF \varphi \equiv E[\text{true} U \varphi]$,

- $AG \varphi \equiv \neg EF \neg \varphi$,

- $AF \varphi \equiv \neg EG \neg \varphi$,

- $A[\varphi U \psi] \equiv \neg E[\neg \psi U (\neg \varphi \wedge \neg \psi)] \wedge AF \psi$,

- $A[\varphi R \psi] \equiv \neg E[\neg \varphi U \neg \psi]$,

- $E[\varphi R \psi] \equiv \neg A[\neg \varphi U \neg \psi]$.

CTL — Some Examples

■ Some examples of CTL formulae:

- ▶ Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 - $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- ▶ It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 - $EF (Start \wedge \neg Ready)$
- ▶ Whenever a request occurs (i.e., Req holds), then it will eventually be acknowledged (i.e., Ack will hold).
 - $AG (Req \rightarrow AF Ack)$
- ▶ In any run of the system, $DeviceEnabled$ is true infinitely often.
 - $AG AF DeviceEnabled$
- ▶ From any state, the system can be restarted (i.e., get to a $Restart$ state).
 - $AG EF Restart$
- ▶ The $Reset$ signal is initially set, and from the next state on, it is never set again.
 - $Reset \wedge AX AG \neg Reset$
- ▶ The $Reset$ signal is initially set, but once it is unset, it is never set again.
 - $Reset \wedge AG (\neg Reset \rightarrow AG \neg Reset)$
- ▶ The $AccConn$ signal can be set only after the $StartAcc$ signal arrives.
 - $A[StartAcc R (\neg AccConn)]$

CTL Model Checking

The Basic Idea

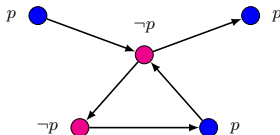
- The **CTL model checking question** to be answered: Given a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP and a CTL formula φ over AP , does $M \models \varphi$ hold?
- A **very basic approach** to answer the CTL model checking question by the so-called **explicit-state model checking**:
 - ▶ For every **subformula** ψ of φ , **label by** ψ all those states s of M in which ψ holds (i.e., $M, s \models \psi$).
 - ▶ Perform the labelling **from the inner-most subformulae** (i.e., the most nested ones) going **to the outer ones** exploiting the already computed labels (with atomic propositions corresponding to the original labels of M).
 - ▶ Check whether each state in S_0 gets labelled by φ .
- It is enough to consider the **basic operators** of CTL, i.e., the below syntax for $p \in AP$:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \vee \varphi \mid EX \varphi \mid E[\varphi U \varphi] \mid EG \varphi.$$

$\text{Label}(\neg\varphi), \text{Label}(\varphi_1 \vee \varphi_2)$

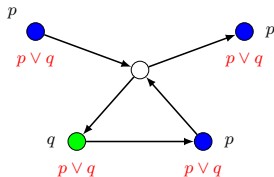
$\text{Label}(\neg\varphi)$

for all $s \in S$ such that $\varphi \notin \text{Label}(s)$ do
 $\text{Label}(s) := \text{Label}(s) \cup \{\neg\varphi\}$



$\text{Label}(\varphi_1 \vee \varphi_2)$

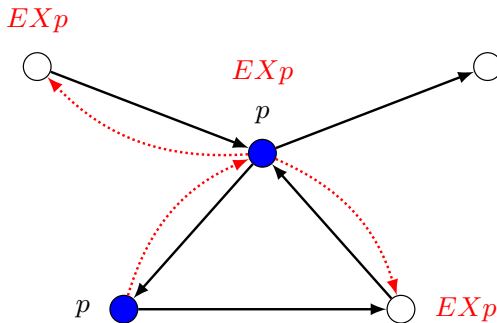
for all $s \in S$ such that $\varphi_1 \notin \text{Label}(s)$ or $\varphi_2 \notin \text{Label}(s)$ do
 $\text{Label}(s) := \text{Label}(s) \cup \{\varphi_1 \vee \varphi_2\}$



Label($EX\varphi$)

Label($EX\varphi$)

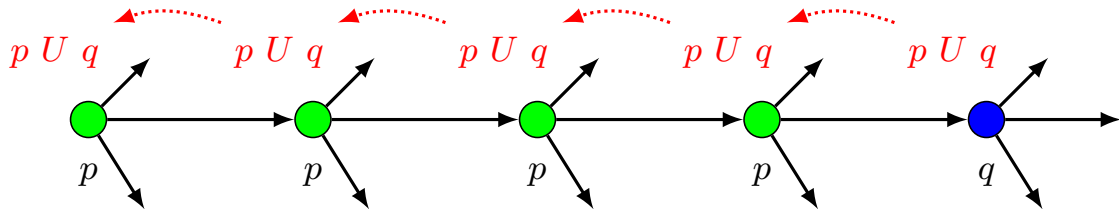
for all $s_2 \in S$ such that $\varphi \in Label(s_2)$ do
 for all $s_1 \in S$ such that $R(s_1, s_2)$ do
 $Label(s_1) := Label(s_1) \cup \{EX\varphi\}$



Label($E[\varphi_1 \text{ U } \varphi_2]$)

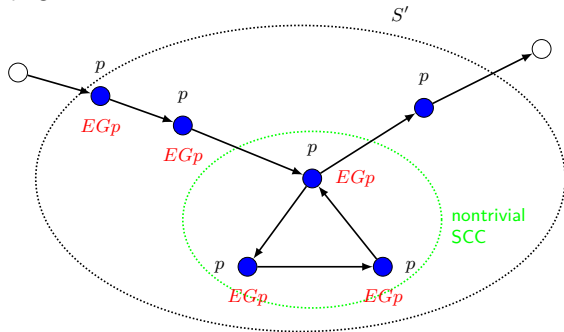
■ The idea:

- ▶ Label first the states already labelled by φ_2 .
- ▶ Look at **predecessors** of states labelled with $\varphi_1 \text{ U } \varphi_2$, and if they are labelled with φ_1 , label them with $\varphi_1 \text{ U } \varphi_2$ as well.



Label($EG\varphi$)

- Based on the following observation: Let $M = (S, S_0, R, L)$ be a Kripke structure, $S' = \{s \in S \mid M, s \models \varphi\}$, and $R' = R \cap (S' \times S')$. For any $s \in S$, $M, s \models EG \varphi$ iff
 - $s \in S'$ and
 - there exists a **path** in the oriented graph $G' = (S', R')$ that leads from s to some node t in a **nontrivial SCC** C of G' .



- An SCC C is **nontrivial** iff either it has more than one node or it contains one node with a self-loop.
- SCCs of a finite oriented graph (V, E) can be computed using **Tarjan's algorithm** in time $O(|E| + |V|)$.

The LTL Logic

LTL — Syntax

- LTL is another sublogic of CTL* that allows only formulae of the form $A \varphi$ in which the only state subformulae are atomic propositions.
- This is, LTL formulae φ are built according to the grammar:
 - ▶ $\varphi ::= A \psi$ (the use of A is often omitted),
 - ▶ $\psi ::= p \mid \neg\psi \mid \psi \vee \psi \mid \psi \wedge \psi \mid X \psi \mid F \psi \mid G \psi \mid \psi U \psi \mid \psi R \psi$ where $p \in AP$.
- Note that LTL speaks about particular paths in a given Kripke structure only—it ignores its branching structure.
- Sometimes, existential LTL allowing formulae of the form $E \varphi$ is used too.
- Note also that while CTL cannot express fairness assumptions (in CTL model checking, they are handled by a special extension of the model checking algorithm), LTL can express fairness assumptions by formulae of the following form:
 - ▶ weak fairness: $(F G \text{ Enabled}) \Rightarrow (G F \text{ Fired})$, i.e., $\Diamond \Box \text{ Enabled} \Rightarrow \Box \Diamond \text{ Fired}$,
 - ▶ strong fairness: $(G F \text{ Enabled}) \Rightarrow (G F \text{ Fired})$, i.e., $\Box \Diamond \text{ Enabled} \Rightarrow \Box \Diamond \text{ Fired}$.

LTL, CTL, and CTL*

- LTL and CTL have an incomparable power:

- ▶ CTL cannot express, e.g., the LTL formula $A (FG p)$,
- ▶ LTL cannot express, e.g., the CTL formula $AG (EF p)$.

- CTL* is strictly more powerful than both LTL and CTL:

- ▶ the disjunction of the above formulae, i.e., $(A (FG p)) \vee (AG (EF p))$, is not expressible in CTL nor LTL.

- To complete the picture, here are the complexities of the appropriate model checking algorithms (we will discuss LTL model checking later on):

- ▶ CTL: linear in $|M|$ and linear in $|\varphi|$,
- ▶ LTL and CTL*: linear in $|M|$ and PSPACE-complete in $|\varphi|$,

where $|M| = |S| + |R|$ and $|\varphi|$ is the number of subformulae of φ .

- Finally, as an example of a logic more general than CTL*, we can mention modal μ -calculus based on least/greatest fixpoint operators on sets of states (basically allowing one to define new, specialised modalities).